IMO Training – Number Theory
IMO Shortlist Problems

Longtin Chan
longtinclt@gmail.com

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Notes

1. (IMO 2010 N1)
   Find the least $n \in \mathbb{N}$ for which $\exists$ a set $\{s_1, s_2, \ldots, s_n\}$ consisting of n distinct positive integers, such that

$$\left(1 - \frac{1}{s_1}\right)\left(1 - \frac{1}{s_2}\right)\cdots\left(1 - \frac{1}{s_n}\right) = \frac{51}{2010}.$$

2. (IMO 2011 N1)
   For any $d \in \mathbb{N}$, let $f(d)$ be the smallest possible integer that has exactly d positive divisors (e.g. $f(1) = 1, f(5) = 16, f(6) = 12$). Prove that for every $k \in \mathbb{Z}_{\geq 0}$, $f(2^k) \mid f(2^{k+1})$.

3. (IMO 2012 N2)
   Find all triples (x, y, z) of positive integers such that $x \leq y \leq z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2).$$

4. (IMO 2013 N1)
   Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$m^2 + f(n) \mid mf(m) + n.$$

5. (IMO 2014 N2)
   Find all pairs (x, y) of positve integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1.$$

6. (IMO 2015 N1)
Find all $M \in \mathbb{N}$ such that the sequence $(a_n)_{n\geq 0}$ defined by

$$a_0 = M + \frac{1}{2} \text{ and } a_{k+1} = a_k \lfloor a_k \rfloor$$

contains at least one integer term.

7. (IMO 2016 N1)
   For any $k \in \mathbb{N}$, denote the sum of digits of k in base 10 by $S(k)$. Find all polynomials $P(x)$ with integral coefficients such that for any positive integer $n \geq 2016$, $P(n) > 0$ and

$$S(P(n)) = P(S(n)).$$

8. (IMO 2017 N1)
For each integer $a_0 > 1$, define the sequence $(a_n)_{n\geq 0}$ by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise.} \end{cases}$$

Find all values of a_0 such that $\exists\ A$ with $a_n = A$ for infinitely many values of n.

9. (IMO 2018 N1)
   Determine all pairs (a, b) of distinct positive integers such that $\exists k \in \mathbb{N}$ for which the number of divisors of ka and that of kb are equal.

10. (IMO 2019 N2)
Find all $a, b, c \in \mathbb{N}$ such that

$$a^3 + b^3 + c^3 = (abc)^2.$$